Parcel & NG911 Data QA: Sheboygan County

Python and PostGIS run side by side on the same county data

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Summary

The toolkit checks **63,391 tax parcels**, **61,811 active address points** and **9,852 road centerline segments** from Sheboygan County's public GIS services for 19 kinds of error that matter to land records and NG911 call routing. It found **1,323 items to review**.

The same 19 checks are written twice: once in Python (GeoPandas) and once as spatial SQL views in PostgreSQL 18.6 with PostGIS 3.6.2. On this run the two versions agree on every check: **1,323 findings in Python, 1,323 in SQL**.

Results by check

Check	What it finds	Severity	SQL	Python	Match	Per 1,000
A01	Missing required attribute	medium	0	0	yes	0.00
A02	Outside its parcel	high	0	0	yes	0.00
A03	Outside every parcel	medium	21	21	yes	0.34
A04	Road link broken	high	0	0	yes	0.00
A05	Street name mismatch	high	28	28	yes	0.45
A06	Number outside road range	high	558	558	yes	9.03
A07	Wrong side of the road	medium	201	201	yes	3.25
A08	Far from its road	medium	47	47	yes	0.76
A09	Duplicate address	high	0	0	yes	0.00
A10	Municipality mismatch	medium	3	3	yes	0.05
P01	Overlapping parcels	high	30	30	yes	0.47

Check	What it finds	Severity	SQL	Python	Match	Per 1,000
P02	Gap between parcels	medium	92	92	yes	1.45
P03	Duplicate parcel ID	high	0	0	yes	0.00
P04	Missing required attribute	medium	138	138	yes	2.18
P05	Sliver parcel	low	125	125	yes	1.97
P06	Invalid geometry	high	5	5	yes	0.08
R01	Mixed parity range	medium	17	17	yes	1.73
R02	Overlapping ranges	high	43	43	yes	4.36
R03	Missing range	low	15	15	yes	1.52

Rates are per 1,000 features of the layer the check looks at (parcels, address points or road segments). Checks with zero findings are kept in the table: a clean result is a result.

Where the findings are

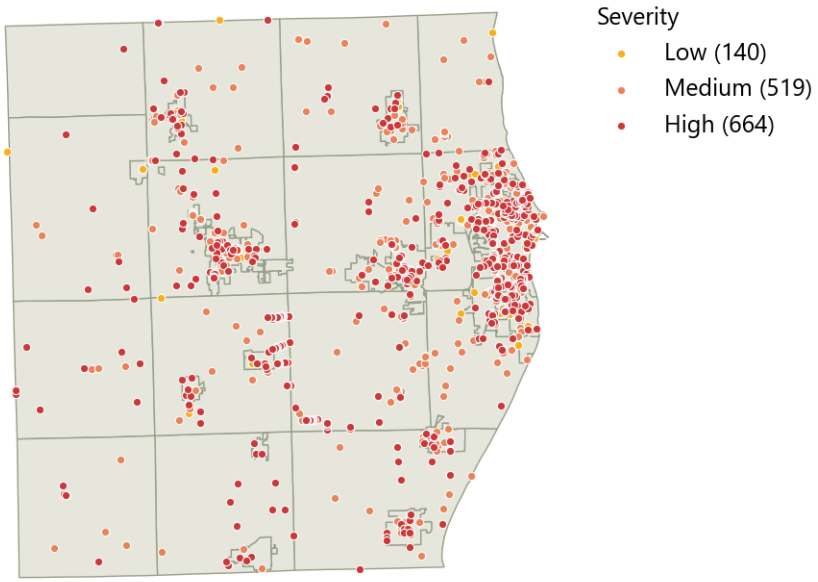


Figure 1: Every finding, colored by severity. Municipal boundaries in gray.

Findings cluster where addresses and small parcels are dense: the City of Sheboygan, Plymouth, Sheboygan Falls and the villages. Rural strings of findings usually follow one road segment whose address range is wrong, which shows up as a run of "number outside road range" findings.

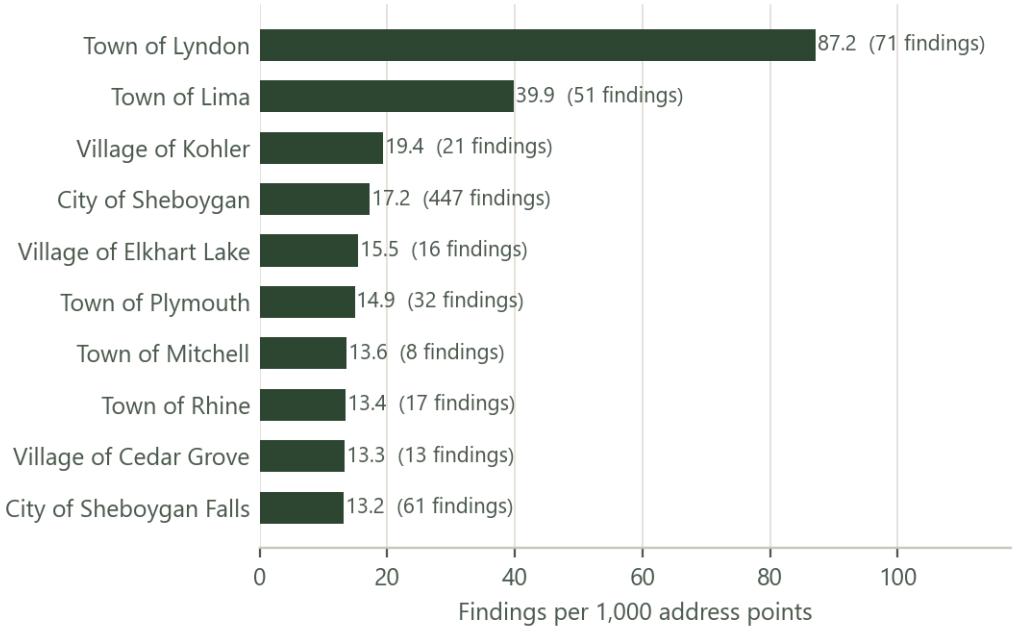


Figure 2: Municipalities with the most address and road findings per 1,000 address points (at least 500 address points).

Per address, two rural towns stand out. In the Town of Lyndon, 39 address points sit outside their road segment's number range and 28 sit unusually far from the segment they are linked to: a handful of segments with wrong ranges or wrong links flag whole strings of rural addresses. The Town of Lima shows the same pattern on a smaller scale.

One example of each

Check	Finding	Feature	Detail
A03	Outside every parcel	SID-19300	Point is not inside any tax parcel
A05	Street name mismatch	SID-10429	Point says 'Indiana Ave', road segment says 'S 23rd St'
A06	Number outside road range	SID-1040	3034 is outside the segment's ranges (3100–3198 this side, 3101–3199 other side); it fits nearby segment RD-1888 (28 m away), so the road link may be wrong
A07	Wrong side of the road	SID-1006	523 is on the left side, but the left range is 500–598
A08	Far from its road	SID-16981	517 m from its road segment
A10	Municipality mismatch	SID-23671	Point says 'Town of Sherman', boundary says 'Village of Random Lake'
P01	Overlapping parcels	59002007451 / 59002007554	Parcels 59002007451 and 59002007554 overlap by 613 m²
P02	Gap between parcels		Gap of 1 m² between parcels
P04	Missing required attribute	59002011174	Field 'OWNERNME1' is empty

Check	Finding	Feature	Detail
P05	Sliver parcel	59010127980	Sliver about 1.2 m wide, 119 m²
P06	Invalid geometry	59012156620	Invalid polygon geometry: Ring Self-intersection[677058.66375846 357602.054499958]
R01	Mixed parity range	RD-12574	Left range 4–7 mixes odd and even
R02	Overlapping ranges	RD-10122 / RD-5166	Even ranges 5990–6058 and 6050–6078 overlap
R03	Missing range	RD-10014	Jolliet Trl segment has no address range, but 3 address(es) link to it

Every finding has a point on the map and a sentence that says what to look at. Three patterns explain most of them:

- **Number outside road range** (the largest group) usually means the road segment's range is too short, or the point is linked to the wrong segment. When a nearby segment of the same street does fit the number, the detail says so.
- **Gaps and overlaps between parcels** are a few square meters each: digitizing slivers where neighboring parcels were drawn at different times.
- **Missing attributes** are mostly parcels without an owner or municipality that are not road right-of-way.

How the database is built

Schema	Table	Rows
core	address_points	61,811
core	municipalities	42
core	parcels	63,391
core	road_centerlines	9,852
lkp	address_status	1
lkp	muni_type	3
lkp	qa_check	19
lkp	road_class	7
lkp	road_status	4
lkp	settings	4

- **`src`** holds the four layers exactly as downloaded, reprojected to Wisconsin Transverse Mercator (EPSG:3071, meters) so distance and area thresholds match the Python version.
- **`lkp`** holds lookup tables, the database version of geodatabase domains: address status, road class and status (which roads are expected to carry address ranges), municipality types, the check list and the thresholds.
- **`core`** holds typed, constrained copies: primary keys, foreign keys to the lookups, CHECK constraints on address ranges, typed geometry columns, a repaired geometry next to the original, generated columns for normalized street names, and GiST spatial indexes.
- **`qa`** holds one materialized view per check, all with the same four columns (`check_code, feature_id, detail, geom`), plus `qa.findings` (every finding as one point layer, used by ArcGIS Pro) and `qa.summary`.

How to rerun

```
activate arcgispro-py3-qa
cd ParcelQA_postgis
REM download the layers, load them into PostGIS, run the SQL checks
python postgis\load_postgis.py config\sheboygan.json
REM run the Python checks on the same files
python parcel_qa.py config\sheboygan.json out
REM build this report (HTML and PDF)
set PGPASSWORD=<postgres password>
quarto render report\parcel_qa_report.qmd
```

When the county updates its data, these steps rebuild every number, table and map in this report.

Rendered November 6, 2026 from PostgreSQL 18.6, PostGIS 3.6.2. Data: Sheboygan County GIS public map services.